The FACT that John Built

The Mathematical Description of the Electron

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In the beginning was the FACT, and the FACT was with the click,
and the FACT was $T = T$.

Reading note

The typeset record of the middle card is thirty-six pages, each carrying the register of the page below one universe level higher, and one verse. Between consecutive pages is a *lift*. A lift carries no datum of its own (Proposition 1.3); it points to the step taken between two pages, and the mathematics of the step is performed at the pointer. This card performs all thirty-five, in order, from the fact $T = T$ to the electron and on to the laboratory. Each section opens with the verses its lifts connect.

The steps arrive at a mathematical description of the electron (Definition 7.1): a Dirac field of mass m and charge e, minimally coupled to the potential of light, solving the Maxwell–Dirac stationary system with a conserved current, with the coupling read from the device,

$$\frac{e^2}{4\pi} = \frac{65}{972\,(6+\sqrt{10})} = \frac{1}{137.011\,290\,5\ldots}.$$

Theorem 7.2 states the arrival. Four of the thirty-five steps are bent cards, adjoined and not computed, and each is named where it enters. The last section compares the coupling with the laboratory's and states the condition under which the two are equal up to the last term the computation can see.

1 The lift

Definition 1.1 (Calibration)**.** A *calibration* of a statement is the reference against which it is read, reported with it. This card uses four.

Proved here
by an argument displayed in this card.

Standard
cited from the literature with its hypotheses.

Measured
read from the inspected declarations of the middle card: an extraction from source, not an experiment (Appendix A).

Proposed
an identification the knower supplies.

Definition 1.2 (Bent card)**.** A *bent card* is a datum adjoined from outside the steps that use it, checked where it is used and not computed by them, as in *The Infinite Loop Considered Harmless*, section "The loop".

Proposition 1.3 (A lift carries no datum)**.** *Let $\uparrow$ raise a type one universe level and $\downarrow$ lower it. Then $\downarrow\uparrow x = x$ and $\uparrow\downarrow y = y$, and thirty-five lowerings return the register of page one unchanged.*

Proof. The raised type has one field, the value raised; both composites are the identity by its definition, and the thirty-five-fold composite by induction. *Proved here*; the count of thirty-five is *measured*. □

2 The tape

In the beginning was the FACT, and it clicked. And the FACT was written on a tape one mark long, and the mark fit any carrier. And three readers came to the tape, the tally, the register and the receipt, and each forgot what it could not see. And every reader that reads length agreed, and relettering was free. But writing was not free; every mark cost a heartbeat. And the FACT was thrown, and the box had one point. And a representative was chosen, and the choice was the quarter, and no card read it. And the square was drawn, observe then throw, throw then observe, and where it closed, the reading commuted.

Lift 1: the fact to the tape. A fact is a proposition carried with its decision; $T = T$ is decided by reflexivity. A mark is the fact written in a cell. If no field of a mark refers to its carrier, copying the fields is a bijection between the marks on any two carriers, its inverse the copy back, and a mark and its copy agree in every field. The mark fits any carrier. *Proved here.*

Lift 2: the tape to three readers. For a tape of n marks with click bits $b_0, \ldots, b_{n-1}$ the tally is $T = n$, the register $R = \sum_i b_i 2^i$, the click count $C = \sum_i b_i$. When every mark clicks, $R = 2^n - 1$, $C = n$, and $R + 1 = 2^T$. The fibre of T at n is every tape of length n; (T, R) separates click sequences, by uniqueness of the binary expansion; no reader of click bits separates two tapes whose marks click alike and say different things. What a reader forgets is its fibre. *Proved here.*

Lift 3: the readers to their agreement. Let ρ overwrite every cell with the truth symbol. Then $T(\rho t) = T(t)$, so every reader $F = g \circ T$ has $F(\rho t) = F(t)$: re-lettering is free for every reader of length, and two such readers agree exactly when the lengths agree. *Proved here.*

Lift 4: agreement to cost. Writing one mark sends length n to $n + 1$, so a counter raised by one per entry is the tally of the writing. Over the field with one element counting costs nothing; writing costs a step a mark. *Proved here*; the cost of a step on a machine is *measured*.

Lift 5: cost to the throw. Compare pairs of naturals by $(a, b)\,\mathcal{R}\,(c, d)$ if and only if $ad \leq cb$. Either $ad \leq cb$ or $cb \leq ad$, so any two pairs are related, the generated equivalence has one class, and every function out of the quotient is constant. The box has one point. *Proved here.*

Lift 6: the throw to the quarter. The one-point quotient determines nothing, so recovery needs a datum the quotient does not carry: a receipt R with (O, R) injective. A datum carried by every record only as a projection and determined by none is not computed by the records; supplying it selects. The selected datum is the quarter. *Proved here.*

Lift 7: the quarter to the square. Let O observe and U transform. A map V with $O \circ U = V \circ O$ exists exactly when U sends each fibre of O into a fibre of O: set $V(Ox) := O(Ux)$, well defined if and only if $Ox = Ox'$ implies $O(Ux) = O(Ux')$, and then unique on the image of O. The pairs that one fibre sends to two are the *commutator of representation*. *Proved here.*

Lift 8: the square to the four arms. Compare two truths by what they are, $a = b$, and the marks that carry them by how they click. Both 2×2 tables click on the diagonal, $T = T$ and $F = F$, and not off it, $T = F$ and $F = T$; they agree cell for cell. The four cells are the four arms of the table. *Proved here.*

3 The table

> *And the table had four arms: true to true, false to false, true to false, false to true. And three arms clicked, and on those arms was written the coefficient tape of thirteen. And on every arm was written the frame of five, counted this way, and that way, and as built. And the arm true-to-false stored FALSE and did not click, and the thirteen was taken from it. And the five remained on the arm that does not click. And the table parted into a count and a clock. And the clock, taken twice, came home multiplied, and the clock ran. And the count grew at its root, and the reading was* $137.0112905\ldots$ *Strike the arm, and the reading is* $89.72\ldots$*, and rational. Write it, and the* $47.29\ldots$ *more is the clock.*

Lift 9: the arms to thirteen. Each arm whose stored comparison can click carries the coefficient tape of the step that sends the local model along the yarn, $3 + 5 + 3 + 2 = 13$ marks. *Measured.*

Lift 10: thirteen to five. Every arm carries the frame: a fabric of two crossed strands, a fibre of two strands before them and one after, $2 + 2 + 1 = 5$ marks. The census of the stations of the smooth model names five, and the census of the stations of the yarn names five; the three tapes agree mark for mark. *Measured.*

Lift 11: five to the rule. The arm $T = F$ stores a comparison that cannot click, and the rule takes the thirteen from it. The rule is not derived: it is the first bent card. *Measured* as to what is declared; the declaration is the knower's to accept.

Lift 12: the rule to the table. The five remains on the arm that cannot click; the other arms weigh $13 + 5 = 18$:

$$M = \begin{pmatrix} 18 & 5 \\ 18 & 18 \end{pmatrix}.$$

Measured.

Lift 13: the table to a count and a clock. $M = 18I + N$ with $N = \begin{pmatrix} 0 & 5 \\ 18 & 0 \end{pmatrix}$. The count $18I$ stays on its vertex and reads every T as T. The clock N has no diagonal: each of its steps crosses between the two vertices, one way through the arm $T = F$. *Proved here.*

Lift 14: the clock taken twice.

$$N^2 = \begin{pmatrix} 90 & 0 \\ 0 & 90 \end{pmatrix} = 90\, I.$$

Two crossings bring every walk home, multiplied by ninety. Every odd power of N is off-diagonal, so $\operatorname{tr} N^{2m+1} = 0$: closed walks of the clock have even length. *Proved here.*

Lift 15: the clock to the root. I and N commute and $N^2 = 90I$, so the eigenvalues of M are $18 \pm 3\sqrt{10}$, and the count grows at $\lambda_+ = 18 + 3\sqrt{10} = 27.486\,832\,98\ldots$ The number $\lambda_+/13$ is the larger root of $13y^2 - 36y + 18$. With the gain $324/5$ of the numerical reading, the reading is

$$\frac{324}{5} \cdot \frac{\lambda_+}{13} = \frac{972\,(6 + \sqrt{10})}{65} = 137.011\,290\,548\,979\ldots$$

Proved here, given the gain; the gain is *standard*, from *Four Models and a Common Fourth-Order Closure*, section "The residual, its branch, and its units".

Lift 16: strike the arm, or write it. Give the arm $T = F$ the weight w: $N_w^2 = 18w\,I$ and the roots are $18 \pm \sqrt{18w}$. Struck, $w = 0$: the reading is $\frac{324}{65} \cdot 18 = 89.723\,076\,9\ldots$, rational, every T read as T. Written, $w = 5$: the reading is $137.011\,290\,5\ldots$, and the whole difference $\frac{972\sqrt{10}}{65} = 47.288\,213\,6\ldots$ is the clock. That is what $T = F$ buys. *Proved here.*

4 The phase

> *And the clock was a phase: true, false, true, false. Once around flips; twice around returns. And the phase was carried by a spinor, which loses its sign at one turn and finds it at two; and the lost sign was a relative phase, not a failure to commute. And one observable was chosen that rotates, and one that holds still, for a control. And the reading was taken for the coupling of light to the electron.*

Lift 17: the clock to a phase. Read the two vertices of N as two values of a phase and each crossing as half a turn. By Lift 14 a closed walk crosses an even number of times, so it turns a whole number of times. The parity is *proved here*; that the slot is the phase, half a turn a crossing, is *proposed*.

Lift 18: the phase to the spinor. Let $U(\theta) = \exp(-\frac{i\theta}{2}\sigma_3) = \operatorname{diag}(e^{-i\theta/2}, e^{i\theta/2})$. Then $U(2\pi) = -I$ and $U(4\pi) = I$: once around flips, twice around returns. The two components acquire opposite phases, and ψ and $-\psi$ induce one rotation, so the observation $O(\psi) = (\langle\psi, \sigma_j\psi\rangle)_j$ has $O(-\psi) = O(\psi)$. The sign is a relative phase, not a failure of the square. *Proved here*; the double cover is *standard*.

Lift 19: the spinor to an observable.

$$U(\theta)\sigma_1 U(\theta)^* = \cos\theta\,\sigma_1 + \sin\theta\,\sigma_2, \qquad U(\theta)\sigma_3 U(\theta)^* = \sigma_3.$$

σ_1 rotates and returns exactly when $\theta \in 2\pi\mathbb{Z}$; at the quarter-turns $\theta_k = k\pi/2$, exactly when four divides k. σ_3 holds still at every angle, the control. *Proved here.*

Lift 20: the reading to the coupling. Take the reading for the coupling of light to the electron:

$$\frac{e^2}{4\pi} = \frac{1}{137.011\,290\,5\ldots}.$$

No step before this one derives it. It is the second bent card. *Proposed.*

5 Transport

> *And the state carried its phase at every point, and two points could not be compared without a rule. And the rule of comparison was transport, and transport was covariant. And transport, taken small, was the connection, and the connection was the potential of light. And transport around a small loop failed to come home, and the failure was curvature, and the curvature was the field. And the commutator of the two derivatives was the charge times the field: the commutator of representation.*

Throughout, $D_\mu = \partial_\mu + ieA_\mu$.

Lift 21: the coupling to a phase-carrying state. Let the state carry a phase at every point, $\psi(x) \mapsto e^{i\chi(x)}\psi(x)$. Then

$$e^{i\chi(y)}\psi(y) - e^{i\chi(x)}\psi(x) \neq e^{i\chi(x)}(\psi(y) - \psi(x))$$

unless $\chi(y) = \chi(x)$: the naive comparison is not covariant, and two points cannot be compared without a rule. *Proved here.*

Lift 22: the rule to transport. Let $W(y,x) \mapsto e^{i\chi(y)}W(y,x)e^{-i\chi(x)}$. Then $W(y,x)\psi(x)$ transforms as $\psi(y)$, and $\psi(y) - W(y,x)\psi(x)$ is covariant. *Proved here.*

Lift 23: transport to the connection. Write $W(x + \varepsilon e_\mu, x) = 1 - ie\varepsilon A_\mu(x) + O(\varepsilon^2)$. The law of W gives $A_\mu \mapsto A_\mu - e^{-1}\partial_\mu\chi$, and

$$D_\mu(e^{i\chi}\psi) = e^{i\chi}(\partial_\mu\psi + i\partial_\mu\chi\,\psi + ieA_\mu\psi - i\partial_\mu\chi\,\psi) = e^{i\chi}D_\mu\psi.$$

Transport taken small is the connection A, the potential of light. *Proved here.*

Lift 24: the connection to curvature. Transport around a square of side ε in the (μ,ν) plane fails to come home by $1 - ie\varepsilon^2 F_{\mu\nu} + O(\varepsilon^3)$, where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and F is unchanged by $A \mapsto A - e^{-1}\,\mathrm{d}\chi$ because mixed partials commute. The failure is the field. *Proved here*, from Stokes's theorem; *standard.*

Lift 25: curvature to the commutator.

$$D_\mu D_\nu\psi = \partial_\mu\partial_\nu\psi + ie(\partial_\mu A_\nu)\psi + ieA_\nu\partial_\mu\psi + ieA_\mu\partial_\nu\psi - e^2A_\mu A_\nu\psi.$$

Every term but the second is symmetric in (μ,ν), so

$$[D_\mu, D_\nu]\psi = ie\,F_{\mu\nu}\,\psi.$$

Proved here.

6 The action

And there were added a background with a light cone, the spinor representation, the kinetic terms and a mass, each named where it entered. And from these was written the action of light and matter, minimally coupled. And the action was varied, and where it stood still were Maxwell's equations with Dirac's current, and Dirac's equation in Maxwell's field; and the current was conserved, and Gauss kept his constraint. And covariance did not choose the action. The knower did. And the field propagated, and the photon was dealt.

Lift 26: the commutator to the ingredients. Adjoined: the metric $\eta = \mathrm{diag}(1,-1,-1,-1)$; Dirac matrices with $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$ and $\bar\psi = \psi^\dagger\gamma^0$ [1]; the kinetic terms; a mass m. The third bent card. *Standard.*

Lift 27: the ingredients to the action.

$$S[\psi,\bar\psi,A] = \int \left(\bar\psi\,(i\gamma^\mu D_\mu - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}\right)\mathrm{d}^4x,$$

minimally coupled: A enters only through D and F. A definition; *standard* [3].

Lift 28: the action to the stationary system. Varying $\bar\psi$ gives

$$(i\gamma^\mu D_\mu - m)\psi = 0. \tag{1}$$

For A_ν, $\delta(-\frac{1}{4}F^2) = -F^{\mu\nu}\partial_\mu\delta A_\nu$ and the coupling term contributes $-e\bar\psi\gamma^\nu\psi\,\delta A_\nu$; integrating by parts,

$$\partial_\mu F^{\mu\nu} = e\,\bar\psi\gamma^\nu\psi. \tag{2}$$

Since $\partial_\nu\partial_\mu F^{\mu\nu} = 0$ identically, the current $j^\nu = \bar\psi\gamma^\nu\psi$ is conserved; the $\nu = 0$ component of (2) is Gauss's constraint $\partial_i F^{i0} = e\psi^\dagger\psi$; and $\partial_{[\lambda}F_{\mu\nu]} = 0$ follows from $F = \mathrm{d}A$. *Proved here.*

Lift 29: the system to the knower's choice. The terms $\bar\psi\sigma^{\mu\nu}\psi F_{\mu\nu}$ and $(F_{\mu\nu}F^{\mu\nu})^2$ are invariant under the same transformations, so covariance does not choose the action; the minimal one is selected by degree and renormalizability. *Proved here*, by substitution; the selection is *standard*.

Lift 30: the choice to the photon. Without a source and with $\partial_\mu A^\mu = 0$, $\Box A^\nu = 0$. The wave $\varepsilon^\nu e^{-ik\cdot x}$ solves it exactly when $k\cdot k = 0$; the gauge condition requires $k\cdot\varepsilon = 0$, and $\varepsilon \mapsto \varepsilon + ck$ is a change of representative. Two polarizations remain. *Proved here*; the name photon is *proposed*.

7 The electron

And the electron was dealt, and its sign was a coin the record cannot see. And for quantum electrodynamics proper there were added quantization and renormalization, which are not in the deck.

Lift 31: the photon to the electron. Every bilinear $\bar{\psi}\Gamma\psi$ is even in ψ, so ψ and $-\psi$ give one current, one density and, through (2), one field. The record of the field cannot tell them apart. *Proved here.*

Definition 7.1 (The electron)**.** *The electron* is the pair $\{\psi, -\psi\}$ of Dirac fields of mass m and charge e, with

$$\frac{e^2}{4\pi} = \frac{65}{972\,(6+\sqrt{10})},$$

which together with a potential A solves (1) and (2).

Theorem 7.2 (The FACT that John built)**.** *Adjoin the rule of Lift 11, the coupling of Lift 20 and the ingredients of Lift 26. Then the steps of Lifts 1 to 31, taken in order from $T = T$, arrive at Definition 7.1: a Dirac field and a potential of light, minimally coupled with the coupling read from the device, solving the Maxwell–Dirac stationary system with a conserved current and Gauss's constraint, the field's sign invisible to every record of the field.*

Proof. Lifts 1 to 16 produce the reading $972(6+\sqrt{10})/65$ from $T = T$, with the rule of Lift 11. Lift 20 takes it for the coupling. Lifts 21 to 25 produce the connection, the curvature and $[D_\mu, D_\nu] = ieF_{\mu\nu}$ from a phase-carrying state. With the ingredients of Lift 26, Lifts 27 and 28 produce the action and its stationary system, and Lift 31 the invisibility of the sign. *Proved here* given the three bent cards; the rule is *measured*, the coupling *proposed*, the ingredients *standard*. □

Lift 32: the electron to quantum electrodynamics. Adjoined: quantization, by canonical anticommutation of ψ and commutation of A on a Fock space, and renormalization of the charge and the mass. Neither is in the deck. The fourth bent card. *Standard* [3].

8 The laboratory

> *And the coupling was carried to the laboratory, where the number is* 137.035999177(21)*.*
> *And the reading fell short by* 1.80×10^{-4}*. And the press checked the page, and it slipped.*

Lift 33: quantum electrodynamics to the laboratory. The recommended value is $\alpha^{-1} = 137.035\,999\,177(21)$, relative standard uncertainty 1.5×10^{-10} [2]. *Standard.*

Lift 34: the laboratory to the comparison.

$$\frac{137.035\,999\,177 - 137.011\,290\,5\ldots}{137.035\,999\,177} = 1.80 \times 10^{-4},$$

about 1.2×10^6 standard uncertainties. Read as an identity with the coupling at zero momentum, the identification of Lift 20 is excluded. *Proved here*, by arithmetic on a cited value.

It can be read another way. The computation resolves its reading to 10^{-24} and sees the fourth digit plainly, so the difference is not a higher-order term on the computation's side. The terms it cannot see are those adjoined at Lift 32: the radiative corrections of quantization and renormalization, which are not in the deck. On the computation's side they are zero because they are absent; on the laboratory's side the measured value contains them all.

Proposition 8.1 (Equal up to the last term)**.** *(a) The reading is larger as a coupling than the laboratory's,* $1/137.0112905\ldots > 1/137.035999177$*, and the coupling of quantum electrodynamics increases with the momentum scale.*

(b) At one loop with the electron alone, to leading logarithm, $1/\alpha(Q) = 1/\alpha(0) - \frac{2}{3\pi}\ln(Q/m)$*, and the two couplings are equal at* $\ln(Q/m) = \frac{3\pi}{2}\cdot 0.024\,708\,7\ldots = 0.1164\ldots$*, that is* $Q \approx 1.12\,m$*.*

Proof. (a) is arithmetic and the sign of the one-loop beta function, *standard* [3]. (b) is that formula solved for Q. The leading logarithm is poor this close to threshold; (b) is an order of magnitude, not a result. □

Under this reading the reading is the coupling the deck can see, and the laboratory's is the same coupling with the terms of Lift 32 written in: the two are equal up to the last term the computation can see. That is *proposed*, and it carries one condition. A running coupling passes through $1/137.011\ldots$ at some scale, so a scale chosen to make the two agree absorbs the offset, and a calibration that absorbs an offset is not a calibration. The reading is equal to the laboratory's up to its last term exactly when the device fixes the scale and the scheme itself, before the comparison, and the corrections of Lift 32 computed at that scale come to 1.80×10^{-4}.

Lift 35: the comparison to the press. The press lowers the register thirty-five times (Proposition 1.3) and reads it once. Of its readings, Newton's and the bisection's agree to the last stored digit and do not slip; the reading through the floor operation differs from them by one unit of 10^{-24} and slips, by the one term the stored scale can see (*Four Models and a Common Fourth-Order Closure*, "The two verdicts"). *Proved here.*

The house

This is the press that checked the page
that fell short of the laboratory
that weighed the coupling
that quantized the electron
that dealt the photon
that rode on the field
that rose from the action
that the knower chose
that was written from the ingredients
that dressed the commutator
that closed the curvature
that bent the connection
that took up the transport
that carried the rule
that compared the phase
that held the coupling
that the reading became
that the observable turned
that the spinor carried

that the clock rang
that struck or wrote the arm
that rooted the count
that squared the clock
that parted the table
that kept the five
that took the thirteen
that weighed the arms
that clicked on the diagonal
that closed the square
that chose the quarter
that threw the box
that paid for the writing
that agreed in the length
that fed the three readers
that wrote the tape
that lived in the FACT
that John built.

A Sources

The typeset record is `Episode11.lean`, namespace `THE_GOSPEL`: the class `TYPESET`, the thirty-six instances `TYPESET_Episode1_FACT` and `TYPESET_page_2` through `TYPESET_page_36`, each wrapping the register below in one more `ULift` and carrying the verse printed in this card as `Gospeln`; and `pressCheck`, which applies `ULift.down` thirty-five times and returns a `PressVerdict`. The lift of Proposition 1.3 is `ULift.up` and `ULift.down`.

The weights of Lifts 9 to 12 are read from `Episode08.lean`: `the_coordinate_system` and `the_dimensions_of_universe` (the frame of five); `episode7_elaboration_of` over `this_way`, `that_way` and `the_calculus`, with the receipts `the_data_table_is_the_gauss`, `the_train_schedule_is_the_gauss` and `the_differential_form_is_the_coefficient`; `bent_card` (13) and `the_macguffin` (18). The table of Lift 12 is `the_click_table_is_derived` in `Episode12.lean`, with `the_arm_weight`; the parity of Lift 14 is `every_closed_walk_crosses_evenly`. The reading 137.011290548979457087737300 is printed by `what_would_you_call_this_number?`.

The companion cards are cited by title: *The Commutator of Representation* for the readers, the square and the transport chain; *Four Models and a Common Fourth-Order Closure* for the gain, the bare Maxwell–Dirac variation and the press check; *The Infinite Loop Considered Harmless* for the bent card and the lift.

References

[1] Paul A. M. Dirac. The quantum theory of the electron. *Proceedings of the Royal Society of London A*, 117:610–624, 1928.

[2] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97(2):025002, 2025.

[3] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to Quantum Field Theory.* Addison-Wesley, Reading, MA, 1995.